

Ques - Given three integer n , a and b , return n^{th} magical number.
Since the answer may be very large return mod $10^9 + 7$.

Soln: Magical no $\rightarrow$ If it is divisible by a or b .

Test case $\rightarrow n=1, a=2, b=3$

out = 2

Approach: We have to find a number such that it is divisible by a or b but not by both.
or

So, we can use inclusion-exclusion principle here.

if $a=4, b=6, n=20$

(let $n=20$)

Divisible by 4 $\rightarrow 20/4 = 5$

Divisible by 6 $\rightarrow 20/6 = 3$

But numbers divisible by both get counted twice (divisible by LCM)

$\text{LCM}(4, 6) = 12$

Divisible by 12 $\rightarrow 20/12 = 1$

$\text{Count}(x) = 5 + 3 - 1 = 7$

~~Search~~ Since $\text{count}(x) = 7 \geq 2 \Rightarrow$ So valid

Since we have to do searching and Binary search can be used.
Search for $\text{count}(x) \geq n$.

Search space $\rightarrow$ smallest possible magical no, $= 1$
largest " " " $= n * \min(a, b)$

Binary search logic

$mid = (low + high) / 2;$

if $(count(mid) < n) \{$

move right $(low = mid + 1)$

else

mid could be ans $(high = mid)$

Keep shrinking until $low == high$.

Low will point to exact n^{th} magical number.

Code: $\text{int } n^{\text{th}} \text{MagicalNumber}(\text{int } n, \text{int } a, \text{int } b) \{$

const int MOD = $1e9 + 7$;

long long low = 1

long long high = $(\text{long long}) n * \min(a, b);$

long long lcm = $(\text{long long}) a * b / \text{gcd}(a, b);$

while $(low < high) \{$

long long mid = $low + (high - low) / 2;$

long long count = $mid / a + mid / b - mid / lcm;$

if $(count < n) \{$

low = $mid + 1;$

$\}$

else high = mid;

$\}$

return low % MOD;

$\}$

Dry Run

$$a = 4$$

$$b = 6$$

$$n = 5$$

Magical nos $\rightarrow 4, 6, 8, 12, 16, 18, \dots$

$$\gcd(4, 6) = 2$$

$$\text{Lcm} = (4 \times 6) / 2 = 12$$

Search Range

$$\text{low} = \min(a, b) = 4$$

$$\text{high} = n \times \min(a, b) = 5 \times 4 = 20$$

BS (Iterations)

① $l = 4$

$$h = 20$$

$$\text{mid} = (4 + 20) / 2 = 12$$

count magical nos. ≤ 12 , count = 4

we need $n = 5$, $5 > 4 \Rightarrow$ move right

$$\text{low} = \text{mid} + 1 = 13$$

② $l = 13, h = 20, \text{mid} = 16$

$$\text{count} = 5$$

$5 = n$ ✓ Enough magical nos.

Try smaller

③ $l = 13, h = 16, \text{mid} = 14$

$$\text{count} = 4$$

$4 < 5$, move right

④ $l = 15, h = 16, \text{mid} = 15$

$$\text{count} = 4$$

move right

$$l = 16$$

$$l = 16, h = 16 \Rightarrow \text{answer} = 16$$